Chapter 7 Maturation

Electrographic activity undergoes serial change from birth through the pediatric age group into adulthood. Interpreting the developing EEG is challenging, but the basic principles of EEG interpretation remain the same for all ages. Neonatal and infantile studies require special accommodations for recording and review. The EEGs of older children and adolescents resemble those of adults. Interpreting neonatal EEGs, especially those of prematurity, requires a level of expertise beyond the scope of most trainees. They will not be discussed here in detail. This chapter is intended as an introductory framework on maturation rather than as a reference for neonatal EEG interpretations, for which the reader is directed to a standard textbook on the subject.

Recording EEGs in Children

The reader should be familiar with a few important aspects of recording EEGs in early life.

Placement

Neonates often require a reduced electrode array due to a smaller head size. At a minimum, Fp1/2, C3/Cz/C4, T7/8, O1/2, and ear lobes (A1/2) or mastoids (M1/2) should be used. Cz (midline) is recommended in neonates, as positive sharp waves typically occur in this channel. The head size in children increases until about 3 years, allowing more electrodes to be added. Older children may be recorded with a full set of electrodes [1].

Parameters

The American Society of Clinical Neurophysiology (ACNS) recommends using a sensitivity of 7 uV/mm and a low-frequency filter between 0.3 and 0.6 Hz (time contrast 0.27–0.53 s) for neonatal records. The adult setting of 1 Hz (time constant 0.16 s) is inadequate for low-voltage fast activities. Children have higher background voltages than adults, so sensitivity may be reduced to 10 or 15 uV/mm as appropriate. The

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standard paper speed is 30 mm/s, but neonatal records are often compressed to 15 mm/s, resulting in a sharper appearance [1,2].

Polygraphy

The sleep–wake cycle is poorly differentiated in neonates. Additional noncerebral parameters called polygraphs are needed to help differentiate physiological states. Polygraphs are also useful for characterizing physiological artifacts and symptoms such as apnea or arrythmia. Standard polygraphs include respiration, heart rate (EKG), eye movements (EOG), and muscle movements (Chin EMG). Synchronous video allows event characterization. Additional measures, such as upper airway exchange and oxygen saturation, are needed in infants with apneic events [1].

Sleep

Efforts should be made to include sleep, as it helps interpret cerebral maturation. Epileptic discharges also accentuate during light sleep. In children, the sleep record is easier to interpret compared to wakefulness as there is less movement artifact. Feeding or sleep deprivation will improve the chances of recording sleep. Sedatives are discouraged, but oral melatonin may occasionally be useful. Standard recordings should be of at least 1 hour duration to obtain a complete sleep cycle [1]. Figure 7.1 shows a typical neonatal recording with polygraphs.

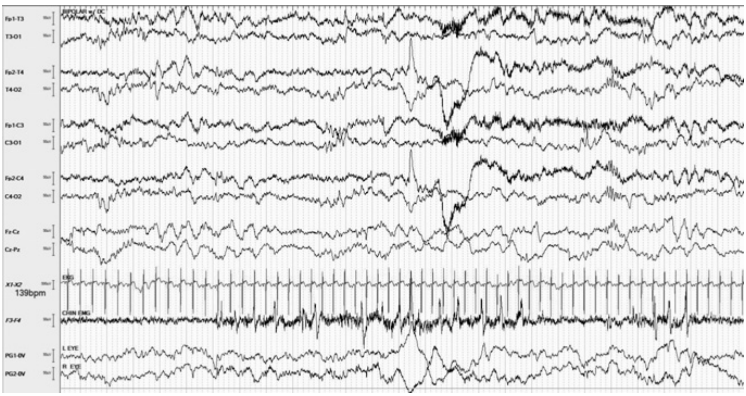

Figure 7.1 Two-day-old boy at term (39 weeks PMA) with twitching, showing the typical channels of a neonatal record (sensitivity 15 uV).

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Basics of Interpretation

The EEG in neonates should be interpreted in the context of two key reference points. These are postmenstrual age (PMA) and physiological state (awake or asleep). These determine the normal expectations for electrographic activity. Persistent deviations outside of these expectations indicate dysmaturity and consequently a greater risk of abnormal neurological outcomes. The sleep–wake cycle and age-specific electrographic patterns (graphoelements) provide additional information about maturity.

Postmenstrual Age (PMA)

Postmenstrual age is the sum of gestational age at birth (number of weeks from the first day of last menstrual period) and chronological age at time of recording (number of weeks postpartum). *Term* is defined as 37–44 weeks PMA, while less than 37 weeks is *preterm* and 44–48 weeks is *postterm*. Traditionally, conceptual age (CA) has also been used. The CA is shorter than the PMA, as the last menstrual period occurs about two weeks before conception [3].

The normal adult EEG is a culmination of serial age-dependent changes of maturation. Assuming other parameters remain constant, there aren't significant differences in the appearance of EEGs among healthy infants of similar PMA, even though gestational ages may differ. For example, the EEG of a healthy 4-week-old baby born at 36 weeks gestation (PMA 40 weeks) is similar to that of a 2-week-old baby born at 38 weeks gestation (PMA also 40 weeks). Premature births will lack the features of a term birth, but these eventually catch up with time. Initially, appreciable changes occur every two to four weeks, but later, the pace slows. By early childhood (1–3 years), the record begins to resemble adulthood and reaches near-maturity in preschool children (3–5 years), with a few minor transformations thereafter [3,4].

Physiological State (Wakefulness, Active, or Quiet Sleep)

EEGs must be interpreted in the context of physiological state (wakefulness or sleep). This is easier in older children and adults but challenging in neonates, as specific sleep architecture is not developed. As the EEG alone isn't enough to determine state, the reader must rely on polygraphs and observation (bedside or video). A clear physiological state typically lasts for at least a minute. The reader must evaluate if the electrographic features during this time are consistent with those normally expected for that state in the context of the PMA. Wakefulness is characterized by open eyes, and this remains the sole identifying feature before 30 weeks PMA, after which polygraphs including respiration and EMG activity become useful. The

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awake neonate may be active (movements) or quiet (wide-open eyes but not moving). Sleep is classified as active (precursor of rapid eye movement [REM] sleep) and quiet sleep (precursor of slow-wave sleep). The normal sleep–wake cycle itself changes through infancy and childhood, as described below [4].

Principles of Maturation

The reader must estimate if the EEG lies within the normal limits of maturation within the context of the PMA and physiological state. This evaluation is based on the predominant electrographic activity (also called background), the sleep–wake cycle, and graphoelements. Continuity and synchrony form the key components of the background development. As the brain matures, the EEG becomes progressively more continuous and synchronous, along with other features, such as voltage, variability, and reactivity. The sleep–wake cycle begins to resemble adults', and transitional graphoelements disappear.

Continuity

Continuity refers to continuous or uninterrupted electrographic activity. Discontinuous periods have a voltage less than 25 uV (attenuation). In a continuous record, these should not exceed 2 s in duration. Sustained continuity is the hallmark of maturation, while persistent discontinuity indicates dysmaturity. The normal neonatal EEG is discontinuous (*trace discontinu*) irrespective of physiological state before 30 weeks PMA. Thereafter, the record becomes progressively more continuous during wakefulness and active sleep (trace discontinu becomes restricted to quiet sleep). At term, the normal neonatal EEG is nearly continuous in all states including quiet sleep. The EEG in a healthy term neonate in quiet sleep shows a continuous but alternating pattern (trace alternans); this is the mature form of trace discontinu. Trace alternans is replaced by slow waves around 42–46 weeks (slow-wave sleep). Figure 7.2 shows persistent discontinuity (dysmaturity) in a term neonate [3].

Synchrony

The near-simultaneous onset of electrographic activity in both hemispheres is called synchrony. Minor lags are permissible (within 1.5 s), but to be synchronous, the overall majority of electrographic activity should begin simultaneously in both hemispheres. The neonatal EEG is normally synchronous before 30 weeks PMA but becomes asynchronous between 30 and 37 weeks, after which synchrony is again restored. As the normal neonatal EEG is synchronous by term, persistent asynchrony is abnormal [3].

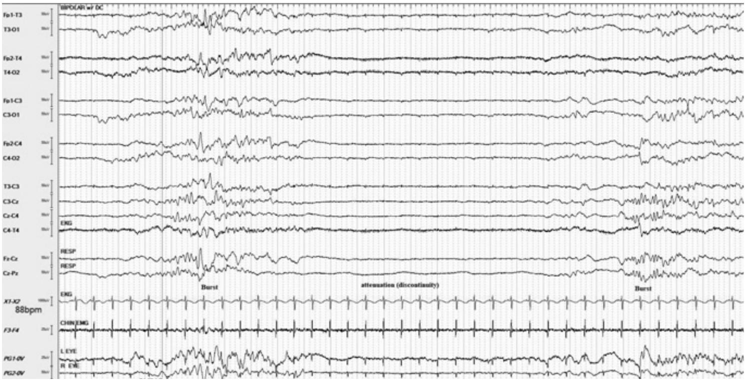

Figure 7.2 Four-day-old girl at term (39 weeks PMA) with neonatal seizures showing an abnormal pattern of persistent discontinuity (sensitivity 15 uV).

Voltage

Voltage is the amplitude or height of the electrographic activity. There is no consensus on the normal limits of voltage in prematurity, but healthy term neonates have voltage greater than 25 uV in all physiological states. Attenuation refers to voltages between 25 and 10 uV, while suppression refers to those less than 10 uV. Suppression suggests severe neuronal dysfunction or injury. The absence of electrographic activity of more than 2 uV when reviewed at a sensitivity of 2 uV/mm is called electrocerebral inactivity (ECI). Several extracerebral conditions, such as poor electrode placement, scalp swelling, and subdural hemorrhage, may also artificially reduce the voltage [3].

Variability and Reactivity

Variability refers to spontaneous changes in the electrographic activity, such as those occurring with state changes (sleep–wake cycle). These normally become established at 30 weeks PMA. Reactivity refers to changes in the electrographic activity occurring secondary to an external stimulus. These may be elicited from around 32 weeks PMA [3]. Both variability and reactivity are discussed in greater detail in Chapter 10.

Maturation of the Sleep–Wake Cycle

Preterm neonates show the early signs of sleep–wake cycling prior to 30 weeks PMA. Later, around 40% of sleep is active, 30% is quiet, and 30% is indeterminate,

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with a duration of sleep cycle around 30–50 min until term. Healthy term neonates normally have a complete sleep–wake cycle over 3 hours and a sleep-only cycle (active–transition–quiet) over an hour. For the first 3 months of age, the infant first falls into active sleep (precursor of REM sleep); later, the infant falls into quiet sleep (precursor of slow-wave sleep) and then transitions to active sleep like adults. Early portions of the EEG during quiet sleep may show trace alternans, but this is replaced with continuous high-voltage slow waves (slow-wave sleep) with maturity. At term, about 50% of sleep is active, whereas in adults, about 20% is REM sleep. The duration of each stage of sleep is about 20 min, though it may be longer. Therefore, standard recordings should be at least 1 hour in duration to obtain a complete sleep cycle [3].

Graphoelements

Graphoelements are distinct neonatal waveforms that temporarily appear during development in an age-dependent manner. These patterns are embedded into the electrographic activity. They appear during specific ranges of PMA and then disappear. Early examples of graphoelements include monorhythmic delta, delta brushes, and rhythmic temporal delta activity. Anterior dysthymia and enconches fontanelles typically occur between 32 and 44 weeks PMA. Anterior dysrhythmia consists of symmetric and synchronous delta runs over the frontal region. Enconches fontanelles are broad-pointed diphasic waves with a small initial upward (negative) deflection followed by a large downward (positive) deflection, most prevalent in transitional sleep. Figure 7.3 shows anterior dysrhythmia and enconches fontanelles [3,4].

EEG of Neonates and Preterm Births

Wakefulness

Wakefulness in a healthy term neonate is characterized by wide-open eyes and irregular respirations. Spontaneous limb and body movements may also occur. The corresponding EEG consists of continuous, medium-voltage (25–50 uV) theta and delta with overriding beta activities. This is called *activite moyenne* (French for “medium”). In premature neonates, prior to 30 weeks PMA, the record is normally discontinuous (trace discontinu). Brief portions of the waking record become continuous after 30 weeks PMA. The normal neonatal EEG is nearly continuous during wakefulness by 34 weeks PMA to term [3]. Figure 7.3 shows the normal EEG of wakefulness at term.

Sleep

Neonatal sleep may be active (precursor of REM sleep) or quiet (precursor of slow-wave sleep). Additionally, transitional and indeterminate stages occur.

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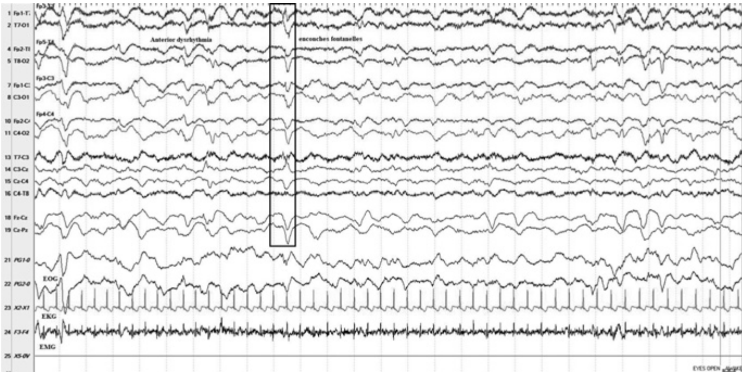

Figure 7.3 Nine-day-old boy at term (40 weeks PMA) during wakefulness showing activite moyenne along with anterior dysrhythmia and enconches fontanelles (sensitivity 15 uV).

During the first 3 months of life, the neonate first falls into active sleep and then transitions to quiet sleep, followed again by active sleep. Later, drowsiness is followed by quiet and then active sleep. This pattern continues in later life, when slow-wave sleep is followed by REM sleep.

Active Sleep

Healthy term neonates in active sleep have their eyes closed, intermittent periods of REM, axial muscular atonia, twitching or jerking movements (myoclonus), and irregular respirations and heart rate. The EEG background shows *activite moyenne*, like that of normal wakefulness. In preterm births prior to 30 weeks PMA, the record in active sleep is discontinuous (trace discontinu), but by 30 weeks, the typical features of active sleep, such as REM, irregular respirations, and body movements, emerge, along with a more continuous EEG. After 34 weeks PMA to term, active sleep in the preterm nearly resembles that of a term neonate with continuous EEG background (activite moyenne). Figure 7.4 shows the normal EEG of active sleep at term.

Quiet Sleep

Healthy term neonates in quiet sleep have their eyes closed, with regular respirations and heart rate. Their movements are minimal (occasional sucking or startles), and rapid eye movements are absent. The EEG background may shows *trace alternans* characterized by alternations of higher voltage at 4- to 5-s bursts (50–150 uV) and brief, low-voltage (25–50 uV) interburst intervals.

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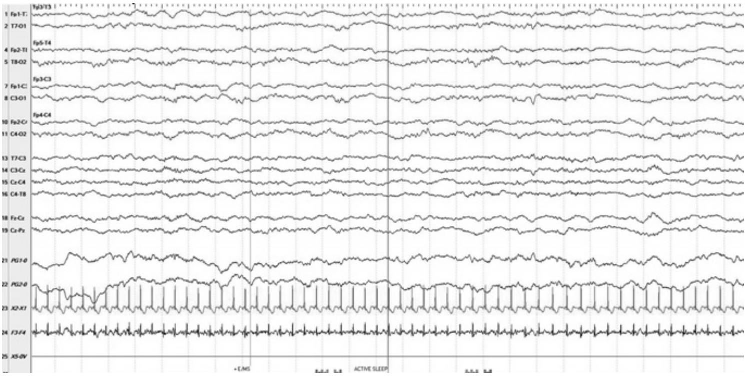

Figure 7.4 Same patient as in Figure 7.3 in active sleep, also showing activite moyenne (sensitivity 15 uV).

The bursts are composed of delta slowing, while there is mixed theta and delta activity during the interburst interval (like activite moyenne). This pattern begins to fade around 42 weeks and disappears by 46 weeks. At this time, quiet sleep is characterized by a continuous higher-voltage, mixed delta and theta background (slow waves). Early sleep spindles (10–12 Hz) first appear at 46 weeks PMA. In preterm neonates, the record is discontinuous irrespective of state prior to 30 weeks PMA. Later, the record becomes progressively more continuous during wakefulness and active sleep but remains discontinuous (trace discontinu) during quiet sleep. Trace discontinu matures into trace alternans at term. Figure 7.5 shows the normal EEG of quiet sleep at term.

Transitional Sleep and Indeterminate Sleep

Temporary stages of incomplete sleep occurring between wakefulness and active and quiet sleep are called transitional sleep. In preterm (especially before 30 weeks), most sleep cannot be properly distinguished into active and quiet sleep, so this is called indeterminate sleep. With an increase in PMA, the periods of indeterminate sleep become limited to transitions between physiological states (transitional sleep) and minimum near term (10%–15%). Excessive amounts of indeterminate sleep in a term neonate may be a sign of abnormality [3].

EEG of Infancy

The EEG in early infancy (first 3 months) is characterized by fading away of neonatal patterns, which are replaced by patterns resembling the electrographic activity of adulthood.

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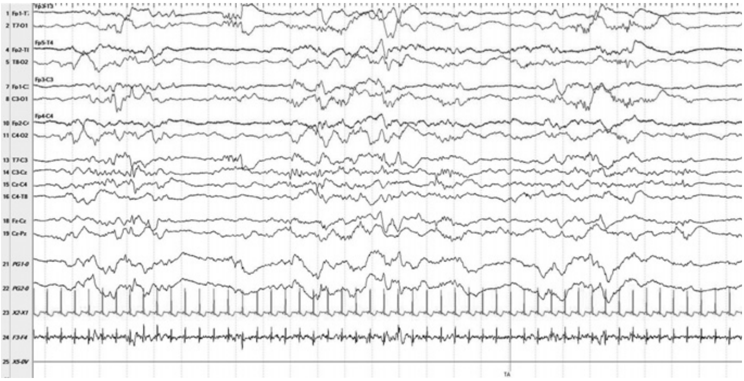

Figure 7.5 Same patient as in Figure 7.3 in quiet sleep, showing trace alternans (sensitivity 15 uV).

Wakefulness

The background frequency begins to transition from the upper delta range (3–4 Hz) to the lower theta range (5–6 Hz) at 3–5 months of age and reaches up to 7 Hz at the end of the first year. Gradually, the emerging posterior-dominant rhythm shows reactivity to eye blink at 3–4 months. Figure 7.6 shows normal EEG of wakefulness in an infant.

Drowsiness

Drowsiness (transition from wakefulness to sleep) is typically characterized by slowing of the electrographic activity to delta range (3–4 Hz). Hypersynchrony refers to generalized bursts of rhythmic, high-amplitude delta (about 4 Hz) that occur during state changes in late infancy (3–12 months) and childhood but sometimes persist in young adults. When hypersynchrony occurs during transition to sleep (drowsiness), it is called hypnogogic hypersynchrony, and when it occurs during arousal from sleep, it is called hypnopompic hypersynchrony. Hypersynchronous bursts may be misidentified as generalized spike wave discharges, especially when faster frequencies are superimposed. However, on close review, the spikes will not be truly associated with after-coming slow waves (more on these later in Chapter 14) [3]. Drowsiness can be easily identified electrographically during transitions after 9 months of age.

Sleep

The trace alternans pattern of neonatal quiet sleep transitions to an occipital-predominant, high-amplitude (150–200 uV), diffuse, polymorphic delta

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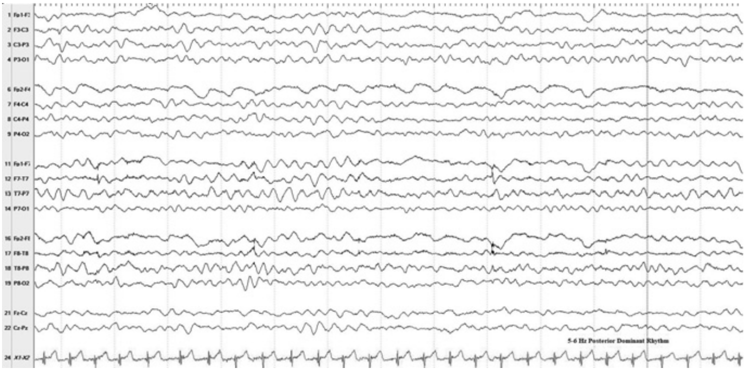

Figure 7.6 Nine-month-old boy with hiccups, 5–6 Hz posterior-dominant rhythm during wakefulness (sensitivity 20 uV). A posterior-dominant rhythm first appears at three months of age.

activity by 42–46 weeks PMA (slow-wave sleep). Asynchronous sleep spindles (12–15 Hz) become apparent over the central regions before 3 months and synchronize after 6 months of age. REM sleep is electrographically characterized by eye movement artifact and diffuse high-voltage delta or theta activities similar to wakefulness. Figure 7.7 shows synchronous sleep spindles.

Sleep–Wake Cycle

The time spent in REM sleep decreases with increasing maturity. It is around 50% at birth and falls to around 40% at 4 months and then 30% after the first year. It is about 20% in adults [4].

EEG of Early Childhood (1–5 Years)

The posterior-dominant rhythm (PDR) attains alpha range (8 Hz) around 2–3 years of age, with some variability. Hypersynchronous bursts decrease in occurrence by around 3 years of age. Drowsiness in children may be heralded by bursts of frontal-predominant theta activity. Stage II (spindle sleep) is now distinctly characterized by synchronous sleep spindles along with prominent vertex waves and K complexes. Stage III (slow-wave sleep) is characterized by posterior-predominant diffuse delta waves, and REM sleep shows diffuse medium-amplitude theta activities. During the preschool years (ages 3–5 years), the normal posterior-dominant rhythm is alpha, though some intermixed theta frequencies are common. Hypnogogic hypersynchrony usually disappears after the age of 3 years. Slow-wave sleep in this age group shows

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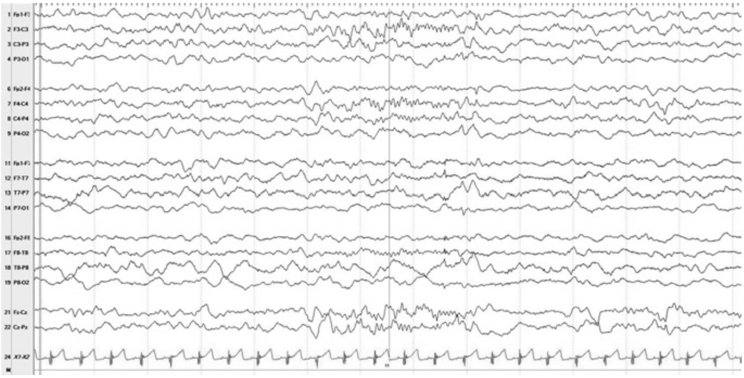

Figure 7.7 Same patient as in Figure 7.6 showing synchronous sleep spindles (sensitivity 20 uV). Asynchronous sleep spindles first appear before three months of age and synchronize around six months of age.

diffuse slow waves with less of an occipital predominance compared to younger children, and REM sleep shows a low-voltage theta background. Preschool children may also cooperate with activation procedures, such as hyperventilation and photic stimulation [4].

EEG of Late Childhood and Adolescence (6–18 Years)

After around 10 years of age, the normal posterior-dominant rhythm during wakefulness is around 10 Hz. Posterior slow waves of youth are common at this age. Hypnogogic hypersynchrony is rarely seen after 6 years of age. A prominent hyperventilation response is common in this age group. Maturation slows during this period, as the EEG closely resembles adulthood [4]. Figure 7.8 shows hypersynchrony during arousal.

Chapter Summary

1. Neonatal EEGs should be interpreted in the context of postmenstrual age (PMA) and physiological state (awake, active, or quite sleep).
2. Sustained continuity is the hallmark of maturation. Early preterm records are discontinuous (trace discontinu) irrespective of state, whereas term records are near-continuous in all states. Between 30 and 37 weeks, the record becomes more continuous during wakefulness and active sleep, but it remains discontinuous during quiet sleep.

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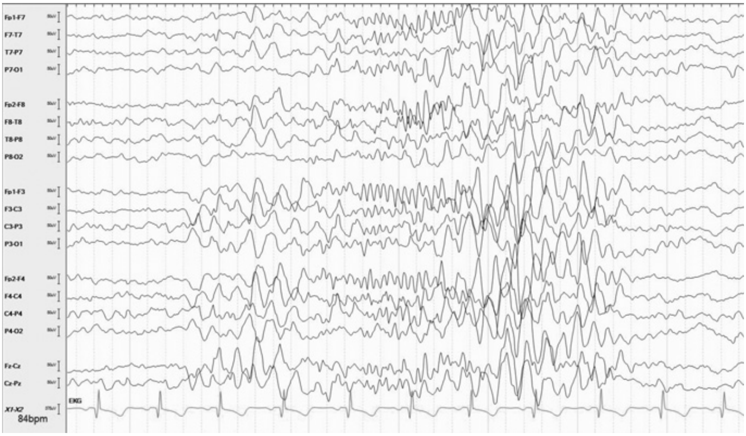

Figure 7.8 Nine-year-old boy with agitation shows hypersynchrony during an arousal (sensitivity 10 uV). Hypersynchrony first appears around three months of age and usually disappears by 3 years of age, though in some children, it may persist into adolescence.

3. Activite moyenne (continuous) is seen during wakefulness and active sleep in a healthy term neonate.
4. Trace alternans is seen during quiet sleep in a healthy term neonate.
5. At term, about 50% of sleep is active, whereas in adults, about 20% is REM sleep.
6. Anterior dysrhythmia and enconches fontanelles are common graphoelements that occur between 32 and 44 weeks PMA.
7. A reactive posterior-dominant rhythm emerges after 3 months of age.
8. Asynchronous sleep spindles appear before 3 months and synchronize by 6 months.
9. The posterior-dominant rhythm attains alpha range by 2–3 years of age.
10. Hypersynchrony appears around 3 months and resolves by 3 years; rarely, it may persist longer. It may be misidentified as generalized spike wave complexes.

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